

1) Divide each term by the previous term to find the common ratio in each of these geometric sequences.

$$5145 \div 735 = 7$$

$$735 \div 105 = 7$$

$$105 \div 15 = 7$$

$$524.88 \div 58.32 = 9$$

$$58.32 \div 6.48 = 9$$

$$6.48 \div 0.72 = 9$$

$$0.72 \div 0.08 = 9$$

$$7 \div 21 = \frac{1}{3}$$

$$21 \div 63 = \frac{1}{3}$$

$$63 \div 189 = \frac{1}{3}$$

$$189 \div 567 = \frac{1}{3}$$

$$\frac{81}{64} \div \frac{27}{32} = \frac{3}{2}$$

$$\frac{9}{16} \div \frac{3}{8} = \frac{3}{2}$$

$$\frac{27}{32} \div \frac{9}{16} = \frac{3}{2}$$

$$\frac{3}{8} \div \frac{1}{4} = \frac{3}{2}$$

2) a) This sequence is geometric. Show that it is and write a sentence to justify why it is.

$$768, 576, 432, 324, 243, \dots$$

$$\begin{array}{cccc} \text{↖} & \text{↖} & \text{↖} & \text{↖} \\ \times \frac{3}{4} & \times \frac{3}{4} & \times \frac{3}{4} & \times \frac{3}{4} \end{array}$$

$$243 \div 324 = \frac{3}{4}$$

$$432 \div 576 = \frac{3}{4}$$

$$324 \div 432 = \frac{3}{4}$$

$$576 \div 768 = \frac{3}{4}$$

The sequence has a common ratio of $\frac{3}{4}$ between successive terms therefore is geometric.

You might also have said...

The sequence has a constant ratio of 4 : 3 between successive terms therefore is geometric. Remember that the common ratio should be given as a single value though.

$$\begin{array}{ccc} \div 192 & \left(\begin{array}{c} 768 \\ 4 \end{array} : \begin{array}{c} 576 \\ 3 \end{array} \right) & \div 192 \end{array}$$

$$\begin{array}{ccc} \div 144 & \left(\begin{array}{c} 576 \\ 4 \end{array} : \begin{array}{c} 432 \\ 3 \end{array} \right) & \div 144 \end{array}$$

$$\begin{array}{ccc} \div 108 & \left(\begin{array}{c} 432 \\ 4 \end{array} : \begin{array}{c} 324 \\ 3 \end{array} \right) & \div 108 \end{array}$$

$$\begin{array}{ccc} \div 81 & \left(\begin{array}{c} 324 \\ 4 \end{array} : \begin{array}{c} 243 \\ 3 \end{array} \right) & \div 81 \end{array}$$

b) This sequence is not geometric. Show that it is not and write a sentence to justify why it is not.

$$756, 252, 126, 42, 21, \dots$$

$$\begin{array}{cccc} \text{↖} & \text{↖} & \text{↖} & \text{↖} \\ \times \frac{1}{3} & \times \frac{1}{2} & \times \frac{1}{3} & \times \frac{1}{2} \end{array}$$

$$21 \div 42 = \frac{1}{2}$$

$$126 \div 252 = \frac{1}{2}$$

$$42 \div 126 = \frac{1}{3}$$

$$252 \div 756 = \frac{1}{3}$$

The sequence has a multiplicative relationship of $\times \frac{1}{3}$ between some terms and $\times \frac{1}{2}$ between others therefore is not geometric.

You might also have said...

The sequence has a ratio of 3 : 1 between some terms and 2 : 1 between others therefore is not geometric.

$$\begin{array}{ccc} \div 252 & \left(\begin{array}{c} 756 \\ 3 \end{array} : \begin{array}{c} 252 \\ 1 \end{array} \right) & \div 252 \end{array}$$

$$\begin{array}{ccc} \div 126 & \left(\begin{array}{c} 252 \\ 2 \end{array} : \begin{array}{c} 126 \\ 1 \end{array} \right) & \div 126 \end{array}$$

$$\begin{array}{ccc} \div 42 & \left(\begin{array}{c} 126 \\ 3 \end{array} : \begin{array}{c} 42 \\ 1 \end{array} \right) & \div 42 \end{array}$$

$$\begin{array}{ccc} \div 21 & \left(\begin{array}{c} 42 \\ 2 \end{array} : \begin{array}{c} 21 \\ 1 \end{array} \right) & \div 21 \end{array}$$



Task B: Using the common ratio

1) Find the missing terms in these geometric sequences.

a) $\left\{\frac{17}{3}, \left[17\right], 51, 153, \left[459\right], \dots\right\}$
 $\times 3 \quad \times 3 \quad \times 3 \quad \times 3$

$$153 \div 51 = 3$$

b) $\left[1125\right], 225, 45, \left[9\right], \left[\frac{9}{5}\right], \dots$
 $\times \frac{1}{5} \quad \times \frac{1}{5} \quad \times \frac{1}{5} \quad \times \frac{1}{5}$

$$45 \div 225 = \frac{1}{5}$$

c) $\left[0.007\right], 0.14, 2.8, \left[56\right], \left[1120\right], \dots$
 $\times 20 \quad \times 20 \quad \times 20 \quad \times 20$

$$2.8 \div 0.14 = 20$$

d) $\left[\frac{7}{3}\right], \left[\frac{14}{9}\right], \frac{28}{27}, \frac{56}{81}, \left[\frac{112}{243}\right], \dots$
 $\times \frac{2}{3} \quad \times \frac{2}{3} \quad \times \frac{2}{3} \quad \times \frac{2}{3}$

$$\frac{56}{81} \div \frac{28}{27} = \frac{2}{3}$$

2) Find the two possible geometric sequences given these missing, non-successive terms.

1 st	2 nd	3 rd	4 th	5 th
4	$\left[20\right]$	100	$\left[500\right]$	$\left[2500\right]$

$\times 5 \quad \times 5 \quad \times 5 \quad \times 5$

$$4 \times r \times r = 100$$

$$4 \times r^2 = 100$$

$$r^2 = 25$$

$$r = 5 \text{ or } -5$$

1 st	2 nd	3 rd	4 th	5 th
4	$\left[-20\right]$	100	$\left[-500\right]$	$\left[2500\right]$

$\times -5 \quad \times -5 \quad \times -5 \quad \times -5$

